

Name: I Forgot Again Team **Date:** 4/5/24 **Class:** Engineering

MyDesign Portfolio

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

Proper Fit: The design should accommodate various car models and measurement configurations to ensure the safety of people and objects.

In our research, we found that a major problem in the previous designs was that they claimed to be a quick fix but it wasn't a one-size-fits-all solution and didn't accommodate all measurements.

We asked multiple drivers for a range of answers and they concluded that the number one priority for our design is it must fit the size/limitations of the car. They said that the prototype must fit the console of their car.

Design Requirement #2 (HIGH Priority)

Durability: The items placed in the prototype must be fully intact during, and after the vehicle is in motion.

When we were researching solutions to the problem, we noticed that some solutions prevented items from falling, but they didn't prevent damage to the objects, as they would often continue to move around eventually scar or damage themselves.

We conducted a survey where we asked 10 drivers about their items. We asked a series of questions in order; "On a scale of 1-10, how often do you drop your items down your car seat gap?", "What items do you regularly drop?" "How much do these items mean to you?" "How would you feel if your items came out damaged, even in the slightest?" We concluded that 90% of day-to-day drivers claim that they not only want to keep their items outside of the gap, but they want their items to be undamaged, and fully optimized after driving.

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Design Requirement #3 (HIGH Priority)

Safety: The driver must be focused on driving and not the location of their objects. In other words, their items must be secure so that the driver can focus.

Other solutions could be uncomfortable because if an item falls out of their pockets and onto their seats, the driver can be distracted and it doesn't fix the root of the problem.

Our primary stakeholders found that when their items fell out onto the seats, they became very distracted and stopped focusing on driving. This tells us that even if we cover the gap, our stakeholders will still be distracted and will cause accidents.

Rubric

Element C: Presentation and Justification of Solution Design Requirements						
C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	No
C2. Requirements list. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are listed, prioritized, clear, and detailed	are listed, prioritized, generally clear, and detailed	are listed, prioritized, generally clear, and somewhat detailed	are listed and prioritized but some or all may be incomplete and/or lack specificity	are listed, formatted, and prioritization is attempted, but these may be partial or overly general	are not present or are too vague
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	Five	Four	Three	Two	One	None

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identified as high priority.						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
C4. Objective & measurable. Design requirements _____.	are consistently objective and measurable	are nearly always objective and measurable	are generally objective and measurable	may be only sometimes objective and/or measurable	are insufficiently measurable	are not measurable
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
C5. Leading to solution. Design requirements _____.	would be highly likely to lead to a tangible and viable problem solution	would be likely to lead to a tangible and viable problem solution	have a potential to lead to a tangible and viable problem solution	have some potential to lead to a tangible and viable problem solution	have little potential to lead to a tangible and viable problem solution	cannot lead to a tangible and viable problem solution
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	no primary stakeholder groups.